GMI German-Malaysian Institute

DIPLOMA PROGRAMME

Practical Test

Academic Period: JANUARY 2026

Program	:	**DIPLOMA IN CYBER SECURITY**
Course	:	**INTRUSION DETECTION AND PREVENTION SYSTEMS**
Code	:	**CIT 0592**
Duration	:	**2 HOURS**
Total Marks	:	**50**

GENERAL INSTRUCTIONS:

1. Write down clearly (with capital letters) your name, trade, group and the date of examination at the bottom of this page.
2. Do not open this question booklet until you are told to do so by the invigilator.
3. Read the specific instructions and questions carefully before answering.

This paper consists of **7** printed pages, including this page.

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Semester/Trade/Group	:	**SEM 4 / DCBS / CBS 1**
Date	:	**11/5/2026**

No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	Installation of Snort and VM/OS	2	2	P3	15			
2.	i. Configure IP address ii. Configure Snort iii. Troubleshooting and Testing	2	3	P3	17			
3.	i. Create icmp rule ii. Create alert rule iii. Troubleshooting and Testing	2	3	P3	8			
4.	Knowledge	3	4	C2	10			
				Total	50			
				Total Marks	/50		/30	
				Signature				

Learning Objectives

- To install Operating System using Linux
- To install, configure and test Snort
- To configure and test Snort Rule

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NO.	CRITERIA	APPROACHES EXPECTATIONS M[1]	MEETS EXPECTATIONS M[2]	EXCELLENT M[3]	MARKS
1	**Installation of Snort and VM/OS**	Unable to install Linux server and Snort 0-5 marks	Able to partially install Linux server and Snort 6-10 marks	Able to install Linux server and Snort with full functionality 11-15 marks	
2	**i. Configure IP address** **ii. Configure Snort** **iii. Troubleshooting and Testing**	Unable to install, configure and test IP address and Snort 0-6 marks	Able to partially install, configure, and test IP address and Snort 7-12 marks	Able to install, configure and test IP address and Snort with full functionality 13-17 marks	
3	**i. Create icmp rule** **ii. Create alert rule** **iii. Troubleshooting and Testing**	Unable to configure and test icmp and alert rule 0-3 marks	Able to partially install, configure, and test icmp and alert rule 4-6 marks	Able to install, configure and test icmp and alert rule with full functionality 7-8 marks	
4	**Knowledge**	Not able to answer questions, not prepared and no confidence at all 0-3 marks	Able to answer questions but with little preparation and confidence 4-7 marks	Able to all answer questions very well and confidently. Very well prepared 8-10 marks	
	TOTAL				

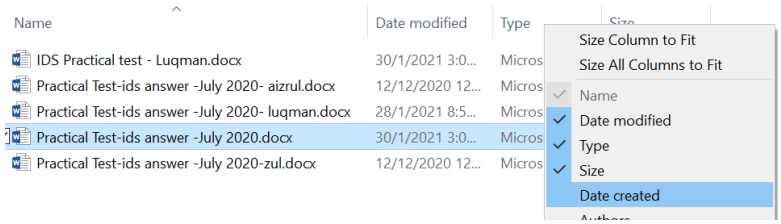
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Create new OS installation and new SNORT installation. (40+10)

1. You need to install and setup snort successfully. You are about to protect **your host** and not using all the rules that currently available by default.

Submission:

a) Attach screen capture the location of your new installed VM folder with 'Date Created' option (right click on the tab above the folder location and tick date created) here. Make sure the name of your VM file and date created should visible.



Example

Your VM Location with Date Created (paste screen capture here)

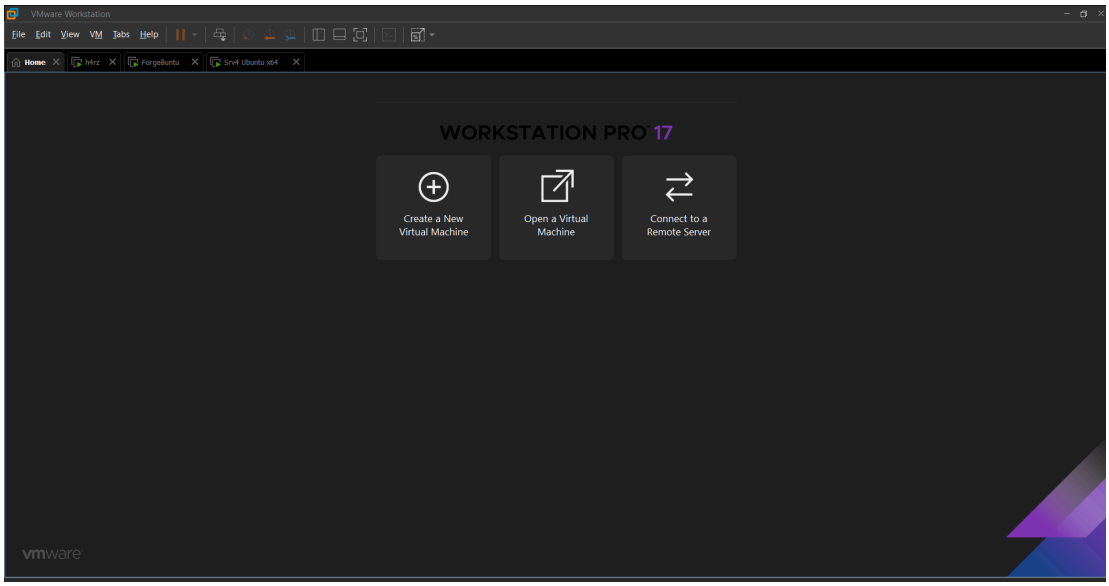


b) Attach your snort.conf upon submission to get marks below.

Task	Marks(32)
1. Install vmware	5
2. Install Snort	10
3. Assign ip address for host in snort conf	2
4. Disable all rules in snort conf	5
5. Able to run snort successfully	10

1. Install VMware

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2. Install Snort

```
┌──(kali㉿kali)-[~]
└─$ snort -v
--------------------------------------------------
o")~   Snort++ 3.12.2.0
--------------------------------------------------
--------------------------------------------------
Network Policy : policy id 1 :
--------------------------------------------------
Inspection Policy : policy id 1 :
--------------------------------------------------
pcap DAQ configured to passive.
--------------------------------------------------
host_cache
    memcap: 33554432 bytes

Snort successfully validated the configuration (with 0 warnings).
o")~   Snort exiting
```

3. Assign IP Address for Host in Snort Conf.

```
-- 1. configure defaults
--------------------------------------------------

-- HOME_NET and EXTERNAL_NET must be set now
-- setup the network addresses you are protecting
HOME_NET = '192.168.58.129/32'
```


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4. Disable all rules in snort conf

```
ips =
{
    -- use this to enable decoder and inspector alerts
    -- enable_builtin_rules = true,
```

5.

```
┌──(kali㉿kali)-[~]
└─$ sudo snort -c /etc/snort/snort.lua -T
--------------------------------------------------
o")~   Snort++ 3.12.2.0
--------------------------------------------------
Loading /etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
```


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2. **Create your own rules to detect activities and produce alert such list below (10 marks):**

Activity	Alert	Marks
A. Ping traffic from vm/physical machine to the snort machine	Somebody is accessing my snort machine	

Details:

Task Activity A	**Marks (8)**
1. Create icmp rule file as test.rule	**2**
2. Create rule **-Rules definition in test.rule** **-alert warning in test.rule** **Establish test.rule in snort conf**	 **2** **2**
3. Test the rule	**2**

Proof of Concept :

Attacker Machine

```
h4rz@ForgeBuntu:~$ ping 192.168.58.129
PING 192.168.58.129 (192.168.58.129) 56(84) bytes of data.
64 bytes from 192.168.58.129: icmp_seq=1 ttl=64 time=4.46 ms
64 bytes from 192.168.58.129: icmp_seq=2 ttl=64 time=0.916 ms
64 bytes from 192.168.58.129: icmp_seq=3 ttl=64 time=0.646 ms
64 bytes from 192.168.58.129: icmp_seq=4 ttl=64 time=1.20 ms
64 bytes from 192.168.58.129: icmp_seq=5 ttl=64 time=0.688 ms
64 bytes from 192.168.58.129: icmp_seq=6 ttl=64 time=1.28 ms
64 bytes from 192.168.58.129: icmp_seq=7 ttl=64 time=2.06 ms
```

Snort Machine

```
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth0
05/11-04:10:22.442740 [**] [1:1000001:1] "Somebody is accessing my snort machine" [**] [Pr
iority: 0] {ICMP} 192.168.58.128 → 192.168.58.129
05/11-04:10:23.442765 [**] [1:1000001:1] "Somebody is accessing my snort machine" [**] [Pr
iority: 0] {ICMP} 192.168.58.128 → 192.168.58.129
05/11-04:10:24.441770 [**] [1:1000001:1] "Somebody is accessing my snort machine" [**] [Pr
iority: 0] {ICMP} 192.168.58.128 → 192.168.58.129
05/11-04:10:25.478181 [**] [1:1000001:1] "Somebody is accessing my snort machine" [**] [Pr
iority: 0] {ICMP} 192.168.58.128 → 192.168.58.129
05/11-04:10:26.477698 [**] [1:1000001:1] "Somebody is accessing my snort machine" [**] [Pr
iority: 0] {ICMP} 192.168.58.128 → 192.168.58.129
```


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Submission:

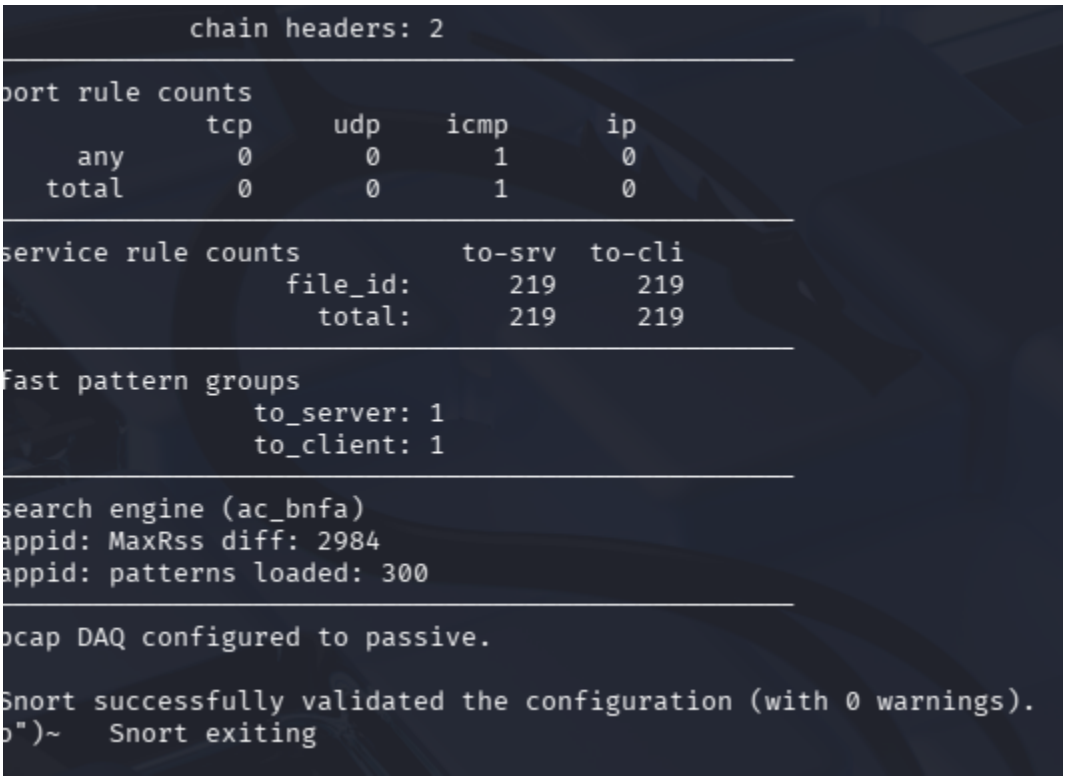
c) Attach test.rule

d) Screen shot the output of command of snort testing below.

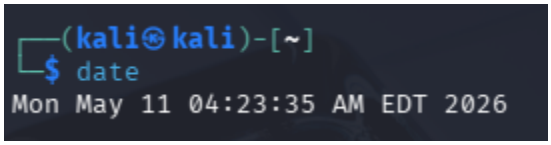
I. /usr/local/bin/snort -T -i eth0 -u snort -g snort -c /etc/snort/snort.conf
(etch0 can be replaced with your own interface)
II. date (to know the exact time of your testing)
III. **Name: yourname (its ok if there is error)**

Output from activity above (paste screen capture here)

i)

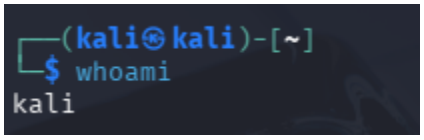


ii)



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iii)



(hariz)

e) **Attach your log file generated from command:**
snort -dve -i eth0 -c/etc/snort/snort.conf -l /var/log/snort/
(etch0 can be replaced with your own interface)

OR
snort -c /etc/snort/snort.conf -K ascii -l /var/log/snort

Location file with date created of output from activity above

(paste screen capture here)

```
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth0
05/11-04:28:46.953094 [**] [1:1000001:1] "Somebody is accessing my snort machine" [**] [Priorit
y: 0] {ICMP} 192.168.58.128 → 192.168.58.129
00:0C:29:87:92:38 → 00:0C:29:FC:E0:85 type:0×800 len:0×62
192.168.58.128 → 192.168.58.129 ICMP TTL:64 TOS:0×0 ID:11122 IpLen:20 DgmLen:84 DF
Type:8  Code:0  ID:1   Seq:1  ECHO

snort.raw[56]:
- - - - - - - - - - - -  - - - - - - - - - - - -  - - - - - - - -
3F 93 01 6A 00 00 00 00  DA BA 00 00 00 00 00 00  ?..j.... ........
10 11 12 13 14 15 16 17  18 19 1A 1B 1C 1D 1E 1F  ........ ........
20 21 22 23 24 25 26 27  28 29 2A 2B 2C 2D 2E 2F   !"#$%&' ()*+,-./
30 31 32 33 34 35 36 37                           01234567
- - - - - - - - - - - -  - - - - - - - - - - - -  - - - - - - - -
```


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3. Answer these questions (10 Marks):

1. Determine file you need to configure for activity question (2) above? **(2 marks)**

   /etc/snort/snort.lua

2. Where to view alerts and its history? **(2 marks)**

Alerts can be viewed in two places:

1. The Terminal (Console): If running Snort with the -A alert_fast or -A cmg flag without a redirect, alerts appear in real-time on your screen.

2. Log Files: For a permanent history, alerts are stored in text files (typically named alert_fast.txt or alert_full.txt) generated by Snort's alert modules.

3. Determine the path for answer question 3(2) above. **(2marks)**

/var/log/snort/

4. Write an alerts definition provided in your log (according to instruction 2(e) above) and explain (what kind of information you get from these alerts). Form a table (2marks for each alert definition and explanation). **(4 marks)**

Sample Alert String:
05/11/26-16:45:10.123456 [] [1:1000001:1] "Somebody is accessing my snort machine" [] [Priority: 0] {ICMP} 192.168.58.128 -> 192.168.58.129

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Explanation :

{ICMP} -> Protocol: Identifies the type of network traffic (in this case, a Ping or Internet Control Message Protocol).

192.168.58.128 -> .129 -> Source & Destination: Shows the IP address of the "attacker" (the .128 machine) and the "target" (your Snort machine).

Checklist of submission:

1. Question paper with answers and screenshots.
   Screenshots:
   a. **VM Location with Date Created**
   b. Snort testing output
   c. **Log file Location with Date Created**
2. **test.rule** attachment
3. **snort.conf** attachment
4. **log file** attachment

THE END

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